ELEC 231 — Mathematical Methods I for ECE · Fall 2026

Week 2: Initial value problems

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Typed from Prof. Korenberg's Fall 2025 handwritten notes, version 2026-09-13.

This note covers selected content from Chapters 1 and 3 of Murray R. Spiegel's *Applied Differential Equations*, and Chapters 6 and 7 of Nilsson & Riedel's *Electric Circuits* (12th ed.).

Notation.

j	imaginary unit, $\mathrm{j}^2=-1$	$\mathcal{L}\{f(t)\}$	Laplace transform
$\mathrm{e}^{-t/\tau}$	natural exponential	$\mathcal{L}^{-1}\{F(s)\}$	inverse transform
$\frac{\mathrm{d}y}{\mathrm{d}t}$, $\frac{\mathrm{d}^2y}{\mathrm{d}t^2}$	derivatives	$v*h$	convolution
$u(t)$	unit step	$\boldsymbol{x}$, $\mathbf{A}$	vector, matrix
$\delta(t)$	unit impulse	τ	time constant

Remark. Page ① of the Week 2 sheets carries the professor's worked solution of the Week 1 tutorial problem (the RL natural response with $L=8\,\mathrm{mH}$, $R=4\,\Omega$ and $I_s=4\,\mathrm{A}$, giving $\tau=2\,\mathrm{ms}$ and $i(t)=4\mathrm{e}^{-500t}$ A, $t\geq 0$). The solution to Pre-tutorial 1 was discussed here; see the Pre-tutorial 1 solutions, where it is typed in full. It is not lecture material, so it is not repeated in these notes.

1 Inductor

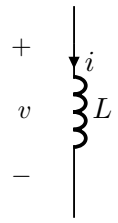


Figure 2.1. The inductor, with the current i entering the + terminal of the voltage v (the passive sign convention used throughout the course).

$$v = L\,\frac{\mathrm{d}i}{\mathrm{d}t},$$

where L — the symbol the professor's arrow points at — is the *inductance*.

In *dc steady state*, all currents and voltages are constant, so then $\frac{\mathrm{d}i}{\mathrm{d}t}=0$ and $v=0$, and the inductor acts as a *short*.

2 Capacitor

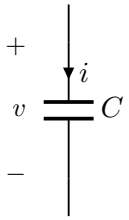


Figure 2.2. The capacitor, drawn with the same sign convention as in Figure 2.1.

$$i = C\,\frac{\mathrm{d}v}{\mathrm{d}t},$$

where C — again the symbol the arrow points at — is the *capacitance*.

In *dc steady state*, all currents and voltages are constant, so then $\frac{\mathrm{d}v}{\mathrm{d}t} = 0$ and $i = 0$, and the capacitor acts like an *open circuit*.

3 Ordinary differential equations (ODE)

Ordinary $\Longrightarrow$ derivatives are with respect to only *one* independent variable, typically time t in this course. For example,

$$\frac{\mathrm{d}i(t)}{\mathrm{d}t} + \frac{R}{L}\,i(t) = 0, \tag{2.1}$$

for the natural response of an RL circuit. Here t is the independent variable and $i(t)$, the inductor current, is the dependent variable.

3.1 Classification

1. **Order.** Note that in Eq. (2.1) the first derivative is the highest derivative that is taken of the *dependent variable*, so the ODE is said to be of 1st order. Later we will see that the natural response of a series RLC circuit satisfies

$$\frac{\mathrm{d}^2 i(t)}{\mathrm{d}t^2} + \frac{R}{L}\,\frac{\mathrm{d}i(t)}{\mathrm{d}t} + \frac{i(t)}{LC} = 0, \tag{2.2}$$

   so this is called a 2nd order equation because $\frac{\mathrm{d}^2 i(t)}{\mathrm{d}t^2}$ is the highest order derivative of the *dependent variable*.

2. **Linear.** An ODE is *linear* if it has form

$$a_n(t)\,\frac{\mathrm{d}^n y}{\mathrm{d}t^n} + a_{n-1}(t)\,\frac{\mathrm{d}^{n-1} y}{\mathrm{d}t^{n-1}} + \cdots + a_1(t)\,\frac{\mathrm{d}y}{\mathrm{d}t} + a_0(t)\,y = f(t), \tag{2.3}$$

   where $a_0, a_1, \ldots, a_n$ are the coefficients.

   Note that if $y(t)$ is replaced by $A\,y(t)$, and similarly for its derivatives, then the left side is multiplied by A. This means that if $y(t)$ is a solution of Eq. (2.3), then $A\,y(t)$ is a solution of the new equation where $f(t)$ is replaced by $A\,f(t)$.

3. **Homogeneous linear ODE:** $f(t) \equiv 0$ in Eq. (2.3).

4. **Constant coefficient linear ODE:** $a_0, a_1, \ldots, a_n$ are constants in Eq. (2.3).

5. **Autonomous ODE:** equation does NOT change with time; equation does NOT depend on the independent variable t.

Example 2.1 (Is the equation a linear ODE?). *Question.* Is

$$t^3 \frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2} + \mathrm{e}^t \frac{\mathrm{d}y(t)}{\mathrm{d}t} + y(t) = 0$$

a linear ODE?

Answer. Yes, because the *dependent variable* $y(t)$ and its derivatives only occur linearly.

Added explanation (not in the 2025 notes). The coefficients t^3 and e^t depend on the independent variable t alone, which is exactly what Eq. (2.3) allows; the equation is linear, but it is not a constant-coefficient one.

4 Natural response of an RC circuit

Source: Nilsson & Riedel (the professor's sub-heading on this page).

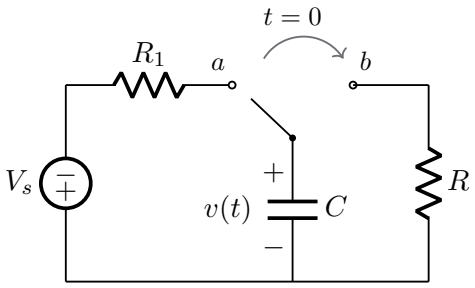


Figure 2.3. Fig. 1: the switch has been in position a for a long time and moves from a to b at $t = 0$.

The switch has been in position a for a long time, so that dc steady-state conditions are reached: capacitors act like open circuits, inductors act like shorts in dc steady state. Then the switch moves from a to b at $t = 0$. Find $v(t)$, $t \geq 0$.

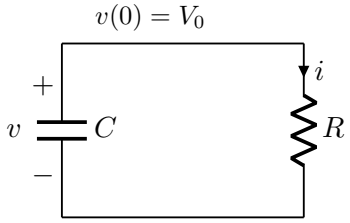


Figure 2.4. Fig. 2: the circuit for $t \geq 0$. Its initial capacitor voltage is the source voltage of Figure 2.3: the shared drawing writes $v(0) = V_0$, and here $V_0 = V_s$.

KVL (Kirchhoff's voltage law), *KCL* (Kirchhoff's current law) applied to the loop of Figure 2.4 give

$$C\,\frac{\mathrm{d}v(t)}{\mathrm{d}t} + \frac{v}{R} = 0, \qquad v(0) = V_s,$$

that is,

$$\frac{\mathrm{d}v(t)}{\mathrm{d}t} + \frac{v}{RC} = 0. \tag{2.4}$$

This equation is analogous to Eq. (2.1) for the natural response of an RL circuit, and Eq. (2.4) has similar solution.

For the above RC circuit, define time constant

$$\tau = RC.$$

Its solution is

$$v(t) = V_s\,\mathrm{e}^{-t/\tau}, \qquad t \geq 0.$$

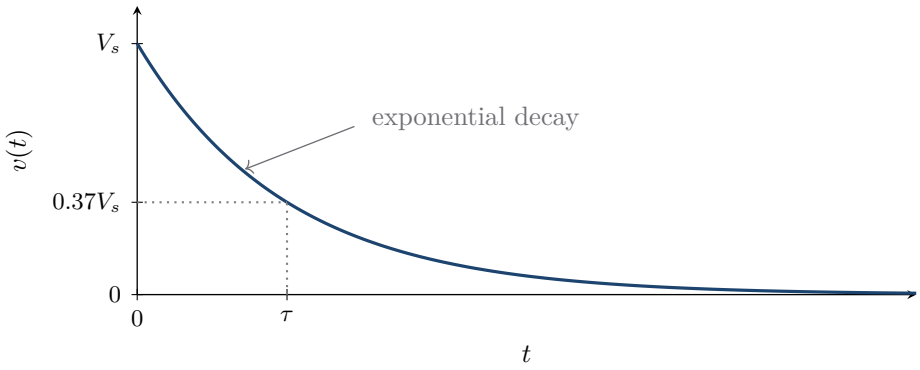


Figure 2.5. The natural response $v(t) = V_s\mathrm{e}^{-t/\tau}$ of the circuit of Figure 2.4, as sketched at the top of page ⑥.

One time constant after the switch has moved to position b, the voltage across the capacitor is $V_s\,\mathrm{e}^{-1} \approx 0.37\,V_s$.

5 Step response of an RC circuit

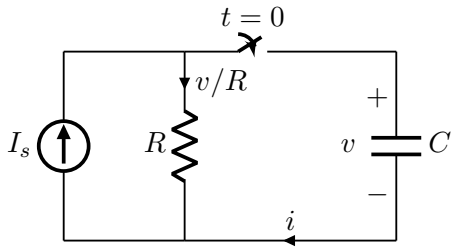


Figure 2.6. The step-response circuit: once the switch closes at $t = 0$ the source I_s, the resistor R and the capacitor C are all in parallel. The current i marked on the bottom conductor is the capacitor current, and v/R is the resistor current.

Kirchhoff's current law at the top node of Figure 2.6 gives

$$C\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{R} = I_s, \qquad t \geq 0,$$
$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{RC} = \frac{I_s}{C},$$

with time constant $= RC$.

Its solution is

$$v(t) = V_f + (V_0 - V_f)\,\mathrm{e}^{-t/\tau},$$

where

$$V_0 = \text{initial value of capacitor voltage},$$
$$V_f = \text{final value of capacitor voltage},$$

so that

$$v(t) = R\,I_s + (V_0 - R\,I_s)\,\mathrm{e}^{-t/RC}, \qquad t \geq 0. \tag{2.5}$$

Added explanation (not in the 2025 notes). The final value in Eq. (2.5) is $V_f = R\,I_s$ because, as $t \to \infty$, the circuit returns to dc steady state: $\frac{\mathrm{d}v}{\mathrm{d}t} \to 0$, so the capacitor current $i = C\,\frac{\mathrm{d}v}{\mathrm{d}t} \to 0$ — the capacitor acts like an open circuit, as in Section 2 — and the whole source current then flows through R, giving $v \to R\,I_s$. Setting $t = 0$ in the same expression gives $v(0) = V_0$, as it must.

The two first-order RC responses. With $\tau = RC$ and $t \geq 0$,

natural response: $v(t) = V_s\,\mathrm{e}^{-t/\tau}$, step response: $v(t) = V_f + (V_0 - V_f)\,\mathrm{e}^{-t/\tau}$,

and for the circuit of Figure 2.6 the final value is $V_f = R\,I_s$.

Remark. The circuit of Figure 2.6 is the subject of Pre-tutorial 2: with $V_0 = 0$, $I_s = 2\,\mathrm{A}$ and $R = 1\,\mathrm{k\Omega}$, what is the capacitor current just before the switch closes compared with just after, and is the capacitor current discontinuous at $t = 0$, unlike the capacitor voltage? That question was posed in the course announcement rather than written on this page; it is stated and solved in full in the Pre-tutorial 2 documents.